HRMS

14 報表分析 — 需求分析書

HR14 REPORTING Service

系統名稱	版本
人力資源暨專案管理系統	1.0
日期	作者
2026-03-17	HRMS 開發團隊

目錄

PM 審查補充

報表分析服務(Reporting & Analytics Service) 需求分析書

版本: 1.0 **所屬領域:** 通用領域 (Generic Domain) **導入階段:** 第二階段

---

1. 服務概述

1.1 核心職責

- CQRS讀模型設計
- 人力資源報表（員工花名冊、差勤統計、離職率）
- 專案管理報表（成本分析、稼動率、獲利率）
- 財務薪資報表（薪資總表、人力成本）
- 客製化儀表板（拖拉式圖表、鑽取功能）
- 資料匯出（Excel/CSV/PDF）

---

2. CQRS架構設計

2.1 讀模型設計

核心概念: Reporting Service維護去正規化的ReadModel，加速查詢

```
寫端（各微服務） → 發布Domain Event
  ↓
Reporting Service訂閱Event → 更新ReadModel
  ↓
查詢API讀取ReadModel → 返回報表數據
```

###2.2 去正規化資料模型範例

EmployeeReportView (員工報表視圖)

```sql
CREATE MATERIALIZED VIEW employee_report_view AS
SELECT
  e.employee_id,
  e.employee_number,
  e.full_name,
  e.hire_date,
  e.employment_status,
  o.organization_name,
  d.department_name,
  m.full_name AS manager_name,
  s.monthly_salary,
  s.payroll_system
FROM employees e
JOIN organizations o ON e.organization_id = o.organization_id
JOIN departments d ON e.department_id = d.department_id
LEFT JOIN employees m ON e.manager_id = m.employee_id
LEFT JOIN salary_structures s ON e.employee_id = s.employee_id AND s.is_active = TRUE;
```

ProjectCostReportView (專案成本報表視圖)

```sql
CREATE TABLE project_cost_snapshots (
  snapshot_id UUID PRIMARY KEY,
  project_id UUID,
  month DATE,

  total_hours DECIMAL,
  total_cost DECIMAL,
  budget_amount DECIMAL,
  budget_utilization DECIMAL,

  employee_costs JSONB, -- [{employeeId, hours, cost}]

  updated_at TIMESTAMP
);
```

更新機制: 訂閱 `TimesheetApproved`, `PayrollRunCompleted` 事件，定期重算

---

3. 核心報表API

3.1 人力資源報表

員工花名冊

```
GET /api/v1/reports/hr/employee-roster
?organizationId={id}&status=ACTIVE

Response 200:
[
  {
    "employeeNumber": "E001",
    "fullName": "張三",
    "department": "研發部",
    "jobTitle": "前端工程師",
    "hireDate": "2025-01-01",
    "serviceYears": 0.9
  }
]
```

差勤統計報表

```
GET /api/v1/reports/hr/attendance-summary
?organizationId={id}&month=2025-11

Response 200:
{
  "month": "2025-11",
  "totalEmployees": 150,
  "averageWorkingHours": 168,
  "totalOvertimeHours": 1200,
  "totalLeaveHours": 500,
  "byDepartment": [...]
}
```

3.2 專案管理報表

專案成本分析

```
GET /api/v1/reports/project/cost-analysis
?projectId={id}&startMonth=2025-01&endMonth=2025-11

Response 200:
{
  "projectId": "uuid",
  "projectName": "XX銀行專案",
  "period": "2025-01 ~ 2025-11",
  "budget": {
    "budgetAmount": 10000000,
    "budgetHours": null
  },
  "actual": {
    "totalHours": 5000,
    "totalCost": 3500000,
    "costUtilization": "35%"
  },
  "monthlyTrend": [
    {
      "month": "2025-01",
      "hours": 400,
      "cost": 280000
    }
  ],
  "profitMargin": "65%"
}
```

稼動率分析

```
GET /api/v1/reports/project/utilization-rate
?departmentId={id}&month=2025-11

Response 200:
{
  "department": "研發部",
  "month": "2025-11",
  "totalEmployees": 50,
  "totalAvailableHours": 8400,
  "billableHours": 6720,
  "utilizationRate": "80%",
  "byEmployee": [...]
}
```

計算公式：

```
稼動率 = 計費工時 / 總工時 × 100%
計費工時 = 專案工時（排除內部專案、訓練等）
```

3.3 財務薪資報表

人力成本分析

```
GET /api/v1/reports/finance/labor-cost
?organizationId={id}&year=2025

Response 200:
{
  "year": 2025,
  "totalLaborCost": 72000000,
  "byMonth": [
    {
      "month": "2025-01",
      "grossWage": 6000000,
      "laborInsurance": 500000,
      "healthInsurance": 300000,
      "pension": 360000,
      "totalCost": 7160000
    }
  ],
  "byDepartment": [...]
}
```

---

4. 客製化儀表板

4.1 Dashboard配置API

建立儀表板

```
POST /api/v1/dashboards
Request:
{
  "dashboardName": "高階主管儀表板",
  "widgets": [
    {
      "widgetType": "KPI_CARD",
      "title": "在職人數",
      "dataSource": "employee_count",
      "position": {"x": 0, "y": 0, "w": 3, "h": 2}
    },
    {
      "widgetType": "LINE_CHART",
      "title": "月度人力成本趨勢",
      "dataSource": "monthly_labor_cost",
      "position": {"x": 3, "y": 0, "w": 9, "h": 4}
    }
  ]
}

Response 201:
{
  "dashboardId": "uuid"
}
```

4.2 關鍵指標

指標	計算方式	數據來源
在職人數	COUNT(status=ACTIVE)	Organization
離職率	離職人數/期初人數×100%	Organization
平均薪資	SUM(netWage)/COUNT	Payroll
人均產值	專案收入/員工數	Project + Payroll
專案獲利率	(收入-成本)/收入	Project + Timesheet + Payroll

5. 資料匯出

5.1 Excel匯出

```
POST /api/v1/reports/export/excel
Request:
{
  "reportType": "EMPLOYEE_ROSTER",
  "filters": {
    "organizationId": "uuid",
    "status": "ACTIVE"
  }
}

Response 200:
{
  "fileUrl": "/downloads/employee-roster-202511.xlsx",
  "expiresAt": "2025-11-25T10:00:00Z"
}
```

5.2 政府申報格式

```
POST /api/v1/reports/export/government
Request:
{
  "formatType": "LABOR_BUREAU_ENROLLMENT",
  "month": "2025-11"
}

Response 200:
{
  "fileUrl": "/downloads/labor-enrollment-202511.xml"
}
```

6. 技術選型

元件	選型
讀模型資料庫	PostgreSQL Materialized View / Elasticsearch
圖表庫	Apache ECharts (前端)
Excel產生	Apache POI
PDF產生	iText
BI工具（可選）	Apache Superset / Metabase

7. 效能優化

7.1 Materialized View定期刷新

```
REFRESH MATERIALIZED VIEW CONCURRENTLY employee_report_view;
```

執行頻率: 每小時

7.2 預計算快照

- 專案成本快照：每日凌晨計算前一日
- 部門人力成本快照：每月結算後計算

文件結束

PM審查補充

報表分析服務 - PM審查補充文件

版本: 1.1
日期: 2025-12-03
補充說明: 補充業務流程圖、循序圖、事件案例等

📋 補充內容

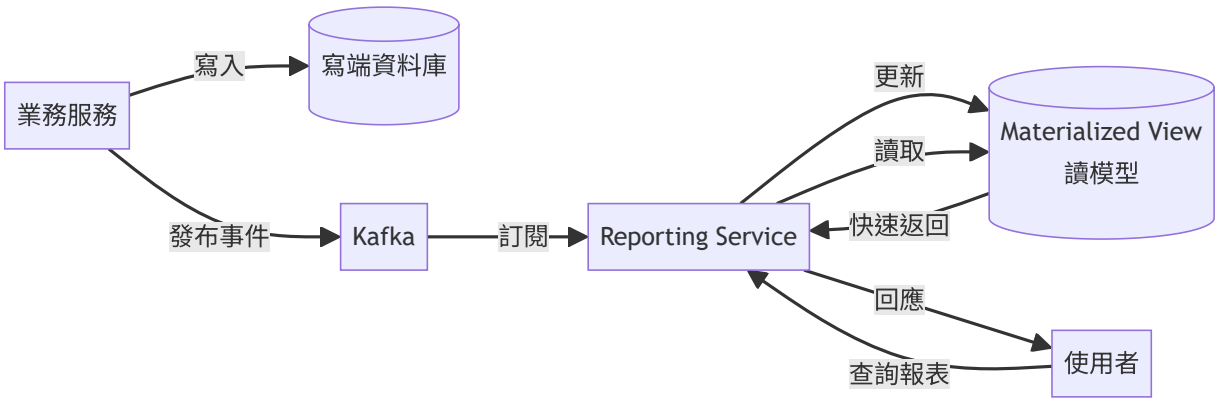
文件增強

- 業務流程圖：CQRS讀模型更新流程
- 循序圖：報表查詢與ReadModel互動
- 業務邏輯：去正規化視圖設計
- 業務案例：高階主管儀表板實例

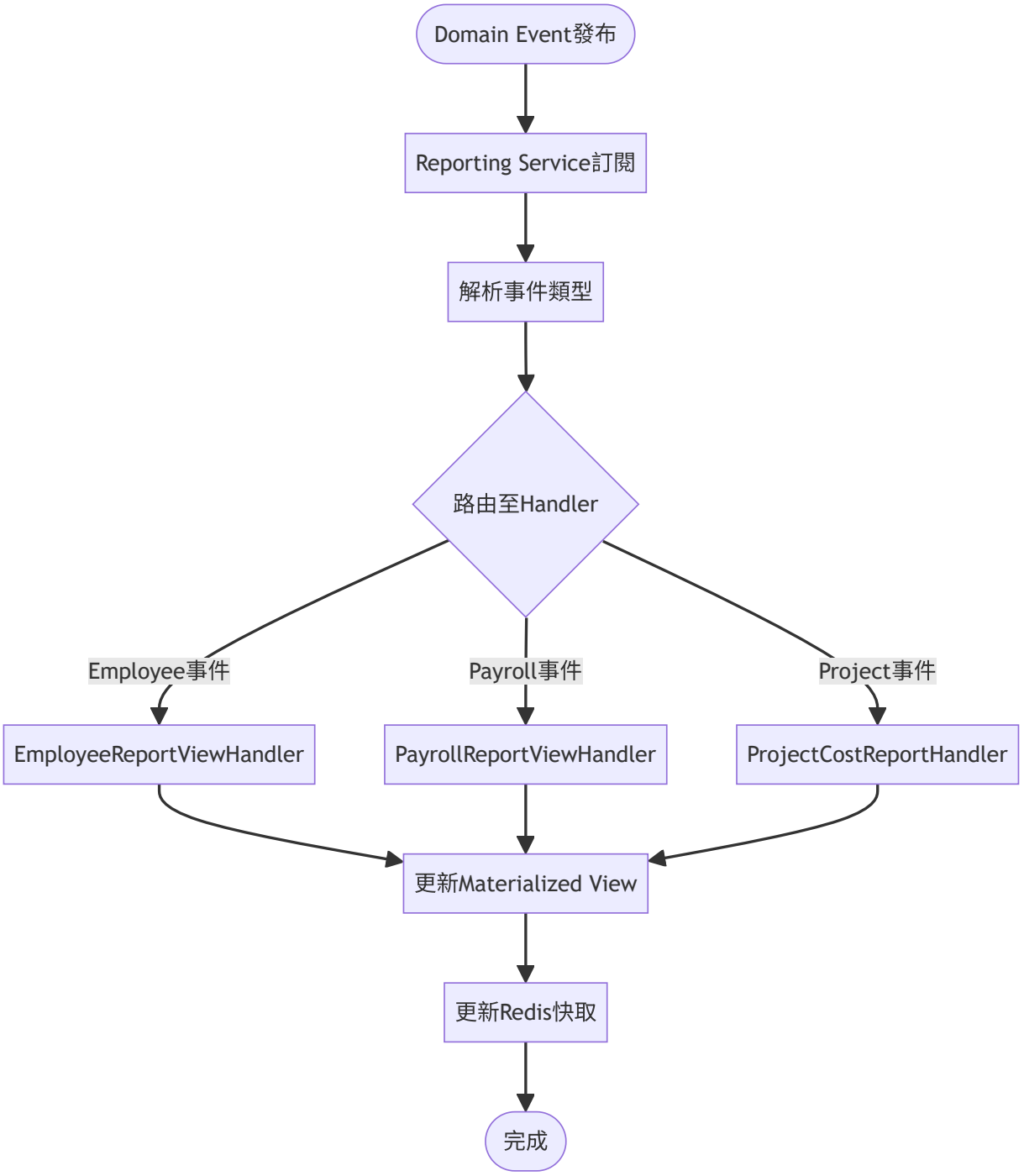
---

1. CQRS架構設計

1.1 讀模型更新流程



1.2 事件驅動ReadModel更新



2. 去正規化視圖設計

2.1 員工報表視圖

```sql
CREATE MATERIALIZED VIEW employee_report_view AS
SELECT
  e.employee_id,
  e.employee_number,
  e.full_name,
  e.national_id_masked,
  e.email,
  e.hire_date,
  e.employment_status,
  e.employment_type,

  -- 組織資訊（去正規化）
  o.organization_name,
  d.department_name,
  d.department_path,  -- 完整部門路徑

  -- 主管資訊
  m.full_name AS manager_name,
  m.email AS manager_email,

  -- 薪資資訊
  s.payroll_system,
  s.monthly_salary,

  -- 計算欄位
  EXTRACT(YEAR FROM AGE(CURRENT_DATE, e.hire_date)) AS service_years,
  CASE
    WHEN e.employment_status = 'ACTIVE' THEN 1
    ELSE 0
  END AS is_active_flag,

  -- 更新時間
  e.updated_at AS last_updated

FROM employees e
LEFT JOIN organizations o ON e.organization_id = o.organization_id
LEFT JOIN departments d ON e.department_id = d.department_id
LEFT JOIN employees m ON e.manager_id = m.employee_id
LEFT JOIN salary_structures s ON e.employee_id = s.employee_id AND s.is_active = TRUE;

-- 建立索引加速查詢
CREATE INDEX idx_emp_report_dept ON employee_report_view(department_name);
CREATE INDEX idx_emp_report_status ON employee_report_view(employment_status);
CREATE INDEX idx_emp_report_hire ON employee_report_view(hire_date);
```

2.2 專案成本快照表

```
CREATE TABLE project_cost_snapshots (
    snapshot_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    project_id UUID NOT NULL,
    snapshot_date DATE NOT NULL,

    -- 專案基本資訊
    project_name VARCHAR(255),
    project_code VARCHAR(50),
    customer_name VARCHAR(255),

    -- 預算資訊
    budget_type VARCHAR(20),
    budget_amount DECIMAL(15,2),
    budget_hours DECIMAL(10,2),

    -- 累計成本（截至snapshot_date）
    total_hours DECIMAL(10,2),
    total_cost DECIMAL(15,2),

    -- 使用率
    cost_utilization_rate DECIMAL(5,4),  -- 成本使用率
    hour_utilization_rate DECIMAL(5,4),  -- 工時使用率

    -- 成員成本明細（JSONB）
    member_costs JSONB,
    /* 範例：
    [
      {
        "employeeId": "uuid",
        "employeeName": "張三",
        "hours": 120,
        "hourlyRate": 250,
        "cost": 30000
      }
    ]
    */

    -- 月度趨勢（JSONB）
    monthly_trend JSONB,
    /* 範例：
    {
      "2025-01": {"hours": 100, "cost": 25000},
      "2025-02": {"hours": 120, "cost": 30000}
    }
    */

    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    UNIQUE(project_id, snapshot_date)
```

```sql
);

-- 每日Job更新昨日快照
CREATE OR REPLACE FUNCTION refresh_project_cost_snapshot()
RETURNS void AS $$
DECLARE
  target_date DATE := CURRENT_DATE - INTERVAL '1 day';
BEGIN
  -- 為所有進行中專案建立/更新快照
  INSERT INTO project_cost_snapshots (
    project_id, snapshot_date, project_name,
    total_hours, total_cost, member_costs
  )
  SELECT
    p.project_id,
    target_date,
    p.project_name,
    -- 計算累計工時與成本...
  FROM projects p
  WHERE p.status = 'ACTIVE'
  ON CONFLICT (project_id, snapshot_date)
  DO UPDATE SET
    total_hours = EXCLUDED.total_hours,
    total_cost = EXCLUDED.total_cost;
END;
$$ LANGUAGE plpgsql;
```

3. 關鍵報表API

3.1 人力盤點報表

```
GET /api/v1/reports/hr/headcount
?asOfDate=2025-12-03&organizationId={id}

Response:
{
  "asOfDate": "2025-12-03",
  "organizationName": "OO科技股份有限公司",

  "summary": {
    "totalEmployees": 150,
    "activeEmployees": 145,
    "onLeaveEmployees": 3,
    "terminatedThisMonth": 2,
    "newHiresThisMonth": 5
  },

  "byDepartment": [
    {
      "departmentName": "研發部",
      "headcount": 50,
      "percentage": "33.3%",
      "avgSalary": 60000,
      "avgServiceYears": 3.5
    },
    {
      "departmentName": "業務部",
      "headcount": 30,
      "percentage": "20%",
      "avgSalary": 55000,
      "avgServiceYears": 4.2
    }
  ],

  "byEmploymentType": {
    "FULL_TIME": 140,
    "CONTRACT": 8,
    "INTERN": 2
  },

  "turnoverRate": {
    "monthly": "1.33%",
    "quarterly": "4.0%",
    "yearly": "15%"
  }
}
```

3.2 部門人力成本分析

```
GET /api/v1/reports/finance/labor-cost-by-department
?year=2025&organizationId={id}

Response:
{
  "year": 2025,
  "organizationName": "OO科技",
  "totalLaborCost": 72000000,

  "departments": [
    {
      "departmentName": "研發部",
      "headcount": 50,
      "totalSalary": 36000000,
      "totalInsurance": 5400000,
      "totalCost": 41400000,
      "percentage": "57.5%",
      "avgCostPerEmployee": 828000
    },
    {
      "departmentName": "業務部",
      "headcount": 30,
      "totalSalary": 18000000,
      "totalInsurance": 2700000,
      "totalCost": 20700000,
      "percentage": "28.75%",
      "avgCostPerEmployee": 690000
    }
  ],

  "monthlyTrend": [
    {
      "month": "2025-01",
      "totalCost": 6000000,
      "salaryPortion": 5000000,
      "insurancePortion": 1000000
    }
  ]
}
```

4. 客製化儀表板

4.1 高階主管儀表板配置

```
{
  "dashboardId": "exec-dashboard-001",
  "dashboardName": "CEO Dashboard",
  "owner": "uuid-ceo",
  "isPublic": false,

  "layout": {
    "columns": 12,
    "rowHeight": 60
  },

  "widgets": [
    {
      "widgetId": "w1",
      "type": "KPI_CARD",
      "title": "在職人數",
      "dataSource": "employee_count",
      "refreshInterval": 3600,
      "position": {"x": 0, "y": 0, "w": 3, "h": 2},
      "style": {
        "icon": "users",
        "color": "#0078D7"
      }
    },
    {
      "widgetId": "w2",
      "type": "KPI_CARD",
      "title": "本月離職率",
      "dataSource": "monthly_turnover_rate",
      "refreshInterval": 3600,
      "position": {"x": 3, "y": 0, "w": 3, "h": 2},
      "threshold": {
        "warning": 2.0,
        "danger": 5.0
      }
    },
    {
      "widgetId": "w3",
      "type": "LINE_CHART",
      "title": "月度人力成本趨勢",
      "dataSource": "monthly_labor_cost",
      "refreshInterval": 86400,
      "position": {"x": 0, "y": 2, "w": 6, "h": 4},
      "chartConfig": {
        "xAxis": "month",
        "yAxis": "cost",
```

```
          "period": "last_12_months"
        }
      },
      {
        "widgetId": "w4",
        "type": "PIE_CHART",
        "title": "部門人數分布",
        "dataSource": "headcount_by_department",
        "refreshInterval": 3600,
        "position": {"x": 6, "y": 2, "w": 3, "h": 4}
      },
      {
        "widgetId": "w5",
        "type": "TABLE",
        "title": "Top 10專案成本",
        "dataSource": "top_projects_by_cost",
        "refreshInterval": 3600,
        "position": {"x": 0, "y": 6, "w": 9, "h": 4},
        "columns": [
          {"field": "projectName", "label": "專案名稱"},
          {"field": "customer", "label": "客戶"},
          {"field": "cost", "label": "成本", "format": "currency"},
          {"field": "budget", "label": "預算", "format": "currency"},
          {"field": "utilizationRate", "label": "使用率", "format": "percent"}
        ]
      }
    ]
  }
```

---

5. 業務案例

業務案例 UC-RPT-001: CEO查看經營儀表板

情境: CEO每日早上查看經營數據

8:30 AM - 登入系統

```
CEO登入 → 進入「經營儀表板」

即時顯示（快取數據，<100ms載入）：

┌──────────────────────────────────┐
│  CEO經營儀表板                     │
├──────────────────────────────────┤
│                                  │
│  ┌────────┐  ┌────────┐         │
│  │在職人數  │  │離職率   │         │
│  │  150人  │  │  1.33% │         │
│  │  ↑+5   │  │  ↓正常  │         │
│  └────────┘  └────────┘         │
│                                  │
│  ┌──────────────────────────┐   │
│  │ 月度人力成本趨勢              │  │
│  │ [折線圖顯示12個月趨勢]        │  │
│  └──────────────────────────┘   │
│                                  │
│  ┌──────────┐  ┌──────────┐    │
│  │部門人數分布 │  │專案成本TOP10│    │
│  │[圓餅圖]   │  │[表格]     │    │
│  └──────────┘  └──────────┘    │
└──────────────────────────────────┘
```

關鍵指標解讀:

```
在職人數：150人
- 較上月+5人
- 趨勢：穩定成長 ✅

離職率：1.33%
- 行業平均：2-3%
- 評價：優於行業水準 ✅

人力成本：
- 本月：600萬
- 年度累計：6,600萬
- 預算控制：良好 ✅

部門分布：
- 研發部：33.3% (50人)
- 業務部：20% (30人)
- 其他：46.7%

專案成本警示：
- XX銀行專案：成本使用率85% ⚠️
- 需關注成本控管
```

8:35 AM - 鑽取分析

```
CEO點擊「XX銀行專案」：
→ 跳轉至專案成本詳細分析頁面

顯示：
- 月度成本趨勢圖
- 成員工時分布
- 預算vs實際對比
- 獲利率預估：15%（原預期25%）

CEO判斷：
- 需與PM討論成本超支原因
- 評估是否需追加預算或調整範圍
```

補充文件結束

主文件: 14_報表分析服務需求分析書.md
修訂日期: 2025-12-03
修訂人: SA

PM 審查補充

工時管理、跨服務整合、支援服務 - PM審查補充文件

版本: 1.1
日期: 2025-12-03
涵蓋服務: 工時管理(07)、績效/招募/訓練(08-10)、簽核流程(11)、通知(12)、文件(13)、報表分析(14)

📋 本文件補充的PM審查項目

CROSS-001: 通知服務完整設計

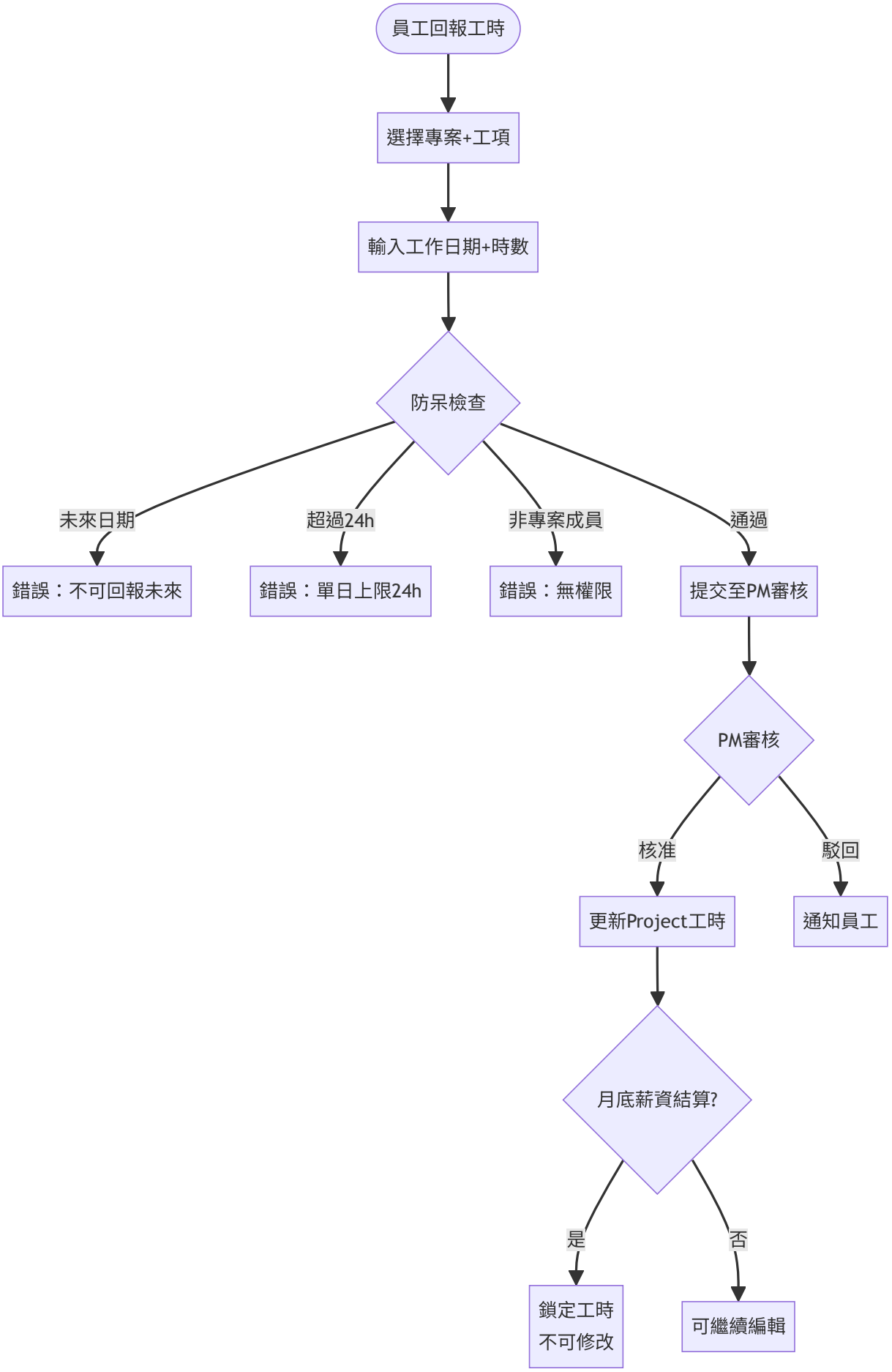
CROSS-002: 資料匯入/匯出功能

CROSS-003: 簽核流程整合設計

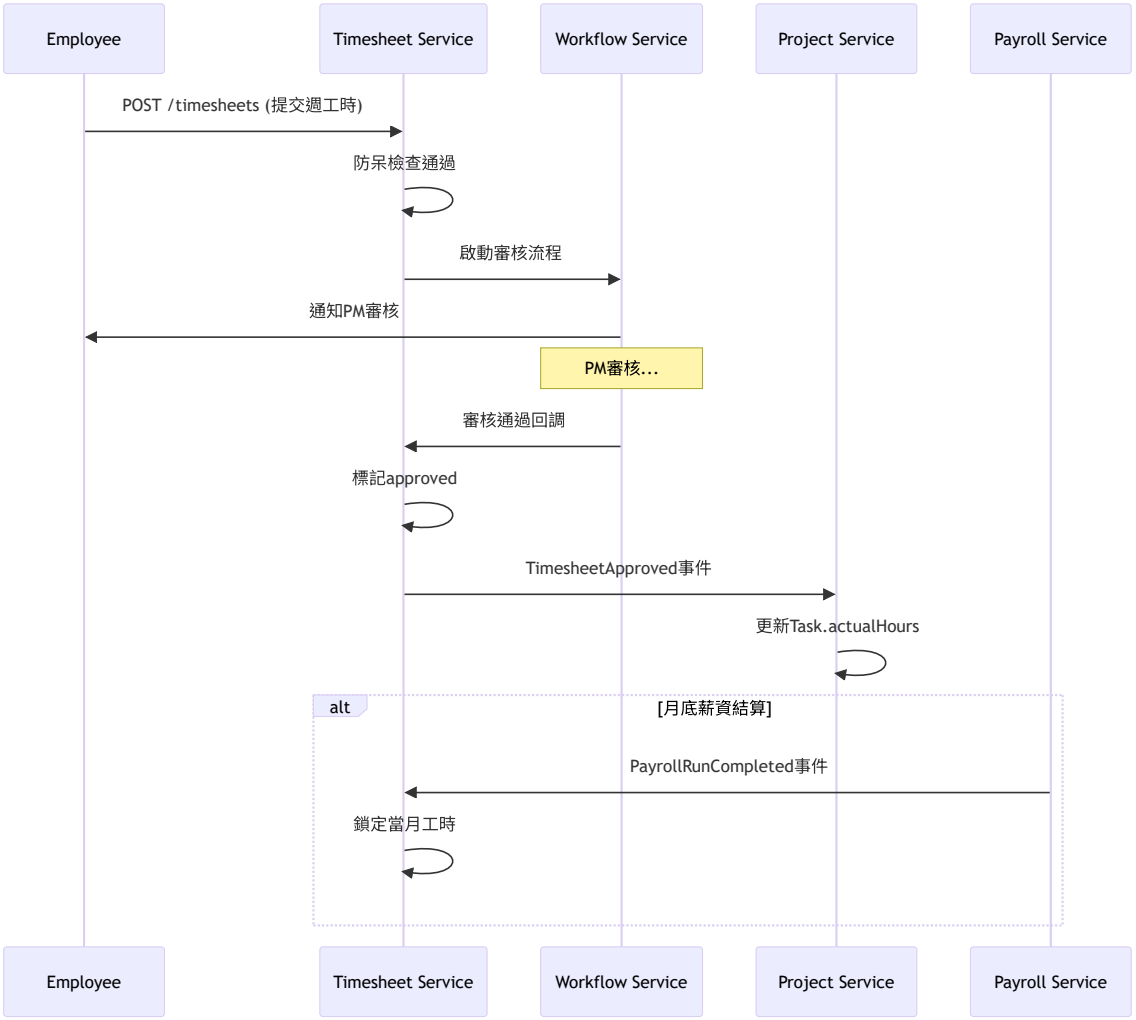
---

1. 工時管理服務 (07)

1.1 業務流程圖：工時回報與鎖定


員工回報工時
選擇專案+工項
輸入工作日期+時數
防呆檢查
未來日期
超過24h
非專案成員
通過
錯誤：不可回報未來
錯誤：單日上限24h
錯誤：無權限
提交至PM審核
PM審核
核准
駁回
更新Project工時
通知員工
月底薪資結算?
是
否
鎖定工時
不可修改
可繼續編輯

1.2 循序圖：工時審核與專案成本更新



1.3 業務案例：專案人力成本追蹤

情境: 追蹤XX銀行專案11月人力成本

```
專案成員工時統計（11月）：
張三 (Tech Lead): 120小時 × 291.67元/h = 35,000元
李四 (Developer): 100小時 × 250元/h = 25,000元
王五 (Developer): 80小時 × 208.33元/h = 16,667元
總計：76,667元

更新Project.actualCost累計：
10月累計：350,000元
11月新增：76,667元
總累計：426,667元
預算使用率：42.67%（總預算1,000,000元）
```

---

2. 簽核流程服務 (11) - CROSS-003

2.1 完整功能設計

代理人機制

```
UserDelegation {
  delegationId: UUID
  delegator: UUID (委託人)
  delegate: UUID (代理人)

  startDate: Date
  endDate: Date

  delegationScope: DelegationScope (ALL, SPECIFIC_TYPES)
  approvalTypes: List<String> (若SPECIFIC_TYPES)

  isActive: Boolean
}

// 範例：主管出國，指定代理人
delegator: 李經理
delegate: 王副理
startDate: 2025-12-10
endDate: 2025-12-20
delegationScope: ALL
```

催辦提醒機制

```
定時Job每日檢查：
- 待辦超過3天 → 發送第一次提醒
- 待辦超過5天 → 發送第二次提醒+通知上級
- 待辦超過7天 → 升級至更高層主管
```

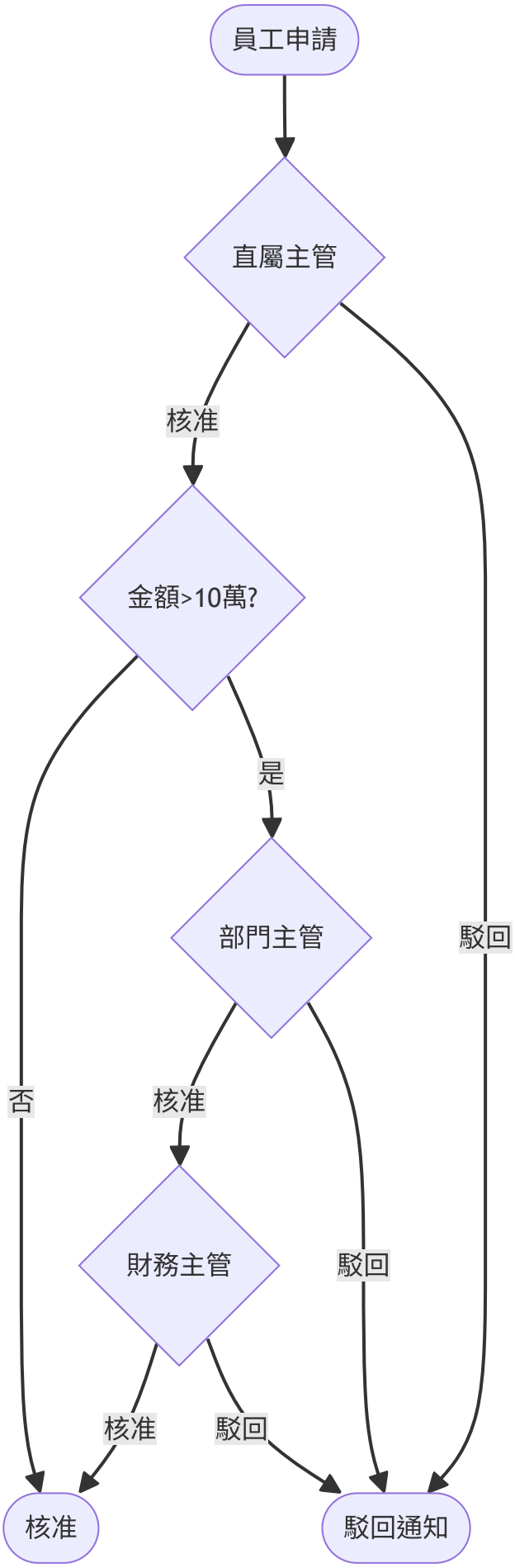
批次簽核

```
POST /api/v1/workflows/tasks/batch-approve
Request: {
  "taskIds": ["uuid1", "uuid2", "uuid3"],
  "comments": "批次核准"
}

Response: {
  "approved": 3,
  "failed": 0
}
```

可視化流程設計器（未來擴充）

- 拖拉式節點設計
- 支援條件分流（if-else邏輯）
- 支援平行會簽（所有人核准/任一人核准）
- 技術選型建議：使用bpmn.js或自行開發React組件

2.2 業務流程圖：多級簽核範例


員工申請
直屬主管
核准
駁回
金額>10萬?
是
否
部門主管
核准
駁回
財務主管
核准
駁回
核准
駁回通知

3. 通知服務 (12) - CROSS-001

3.1 完整架構設計

通知偏好設定

```
UserNotificationPreference {
  userId: UUID

  channels: Map<NotificationType, List<Channel>>
  // 範例：
  // LEAVE_APPROVED → [EMAIL, IN_APP]
  // PAYSLIP_GENERATED → [EMAIL]
  // OVERTIME_ALERT → [EMAIL, PUSH, LINE]

  quietHours: QuietHours (勿擾時段)
  language: String (zh-TW, en-US)
}

QuietHours {
  enabled: Boolean
  startTime: Time (22:00)
  endTime: Time (08:00)
}
```

企業通訊軟體整合

Microsoft Teams整合:

```java
@Service
public class TeamsNotificationService {

    public void sendToChannel(String webhookUrl, String message) {
        MessageCard card = MessageCard.builder()
            .title("系統通知")
            .text(message)
            .themeColor("0078D7")
            .build();

        restTemplate.postForEntity(webhookUrl, card, Void.class);
    }
}
```

LINE Notify整合:

```
POST https://notify-api.line.me/api/notify
Headers:
  Authorization: Bearer {LINE_ACCESS_TOKEN}
Body:
  message=您的請假申請已核准
```

Slack整合:

```
POST https://hooks.slack.com/services/{WEBHOOK_PATH}
Body: {
  "text": "系統通知",
  "blocks": [...]
}
```

3.2 事件訂閱清單（關鍵）

業務事件	通知對象	渠道	優先級
LeaveApplied	主管	Email+系統	NORMAL
LeaveApproved	申請人	Email+推播	NORMAL
OvertimeLimitExceeded	HR+主管	Email+系統	HIGH
PayslipGenerated	員工	Email	NORMAL
PasswordExpiringSoon	使用者	Email+系統	NORMAL
ProjectBudgetAlert	PM+專案總監	Email+Teams	URGENT
InsuranceEnrollmentCompleted	員工	Email	LOW

3.3 通知範本範例

```
【範本：LEAVE_APPROVED】
主旨：請假申請核准通知
內容：
親愛的 {{employeeName}}，

您申請的{{leaveType}}已獲核准：
- 假別：{{leaveTypeName}}
- 期間：{{startDate}} ~ {{endDate}}
- 天數：{{totalDays}}天
- 核准人：{{approverName}}
- 核准時間：{{approvedAt}}

剩餘假期餘額：{{remainingDays}}天

此為系統自動通知，請勿直接回覆。
```

---

4. 資料匯入/匯出 (CROSS-002)

4.1 批次匯入設計

Excel範本格式

```
員工批次匯入範本.xlsx:

Column A: 員工編號* (必填)
Column B: 姓名* (必填)
Column C: 身分證號* (必填，加密儲存)
Column D: Email* (必填)
Column E: 手機
Column F: 部門代碼*
Column G: 職稱*
Column H: 月薪*
Column I: 到職日*
...

備註：
- 標*為必填欄位
- 身分證號格式：A123456789
- Email須唯一
- 部門代碼需先建立
```

匯入API

```
POST /api/v1/employees/import
Content-Type: multipart/form-data

Request:
  file: employees.xlsx
  dryRun: true (預覽模式，不實際匯入)

Response:
{
  "totalRows": 50,
  "validRows": 45,
  "invalidRows": 5,
  "errors": [
    {
      "row": 3,
      "field": "email",
      "error": "Email重複"
    }
  ],
  "preview": [...]
}

若dryRun=false，則實際執行匯入
```

4.2 政府申報格式匯出

勞保局加退保申報XML

```xml
<?xml version="1.0" encoding="UTF-8"?>
<enrollment_report>
  <unit_code>12345678</unit_code>
  <report_month>2025-11</report_month>
  <enrollments>
    <enrollment>
      <id_number>A123456789</id_number>
      <name>張三</name>
      <enrollment_date>2025-11-01</enrollment_date>
      <insured_salary>48200</insured_salary>
    </enrollment>
  </enrollments>
</enrollment_report>
```

4.3 Excel匯出（通用）

```
GET /api/v1/employees/export?format=xlsx&filters=...

Response Headers:
Content-Type: application/vnd.openxmlformats-officedocument.spreadsheetml.sheet
Content-Disposition: attachment; filename="employees_20251203.xlsx"

Body: (Excel binary)
```

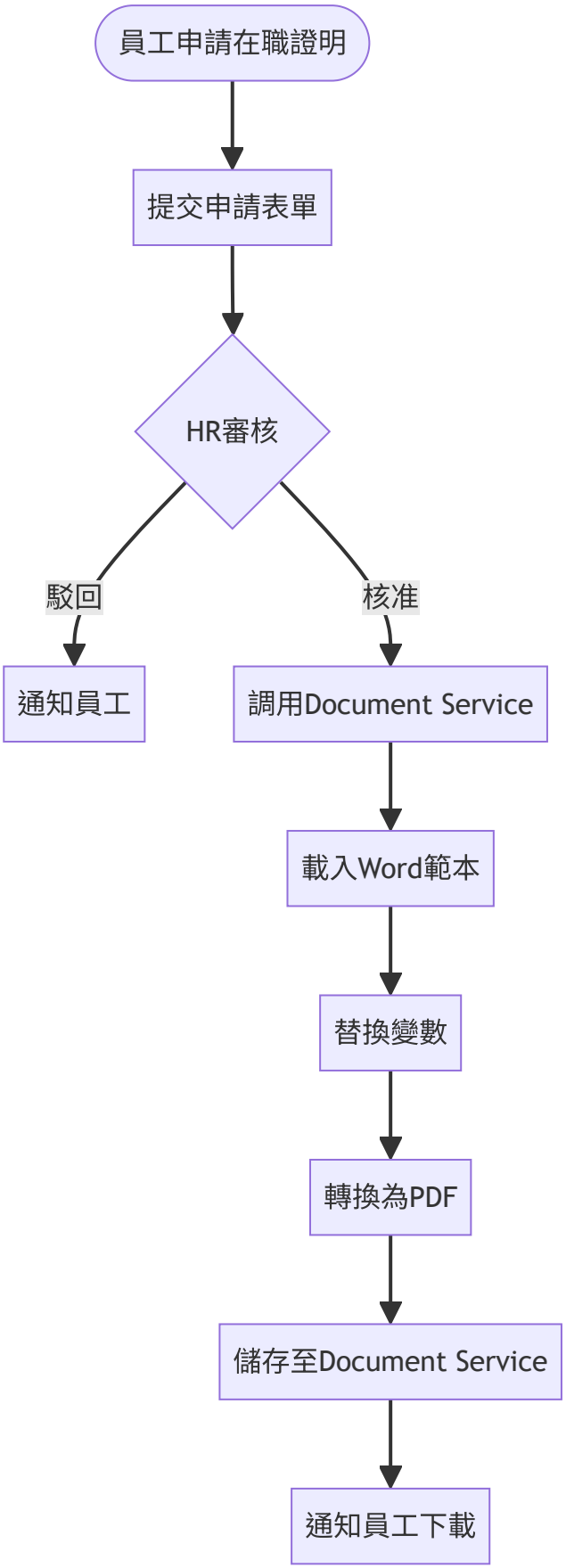
---

5. 文件管理服務 (13)

5.1 文件分類與權限

```
文件類型與可見性：

EMPLOYEE_CONTRACT (員工合約):
  - 擁有者：該員工+HR
  - 儲存位置：/storage/contracts/{employeeId}/

PAYSLIP (薪資單):
  - 擁有者：該員工+財務
  - 加密：AES-256
  - 密碼保護：員工身分證後4碼

EMPLOYEE_PHOTO (員工照片):
  - 擁有者：該員工+全公司（通訊錄）
  - 大小限制：2MB

POLICY_DOCUMENT (政策文件):
  - 擁有者：HR
  - 可見性：全公司
```

5.2 業務流程：產生在職證明



6. 報表分析服務 (14)

6.1 CQRS讀模型更新流程

6.2 關鍵報表API

人力盤點報表

```
GET /api/v1/reports/hr/headcount
Response: {
  "asOfDate": "2025-12-03",
  "totalEmployees": 150,
  "byDepartment": [
    {
      "department": "研發部",
      "count": 50,
      "avgSalary": 60000
    }
  ],
  "byEmploymentType": {
    "FULL_TIME": 140,
    "CONTRACT": 8,
    "INTERN": 2
  },
  "newHires": 5,
  "terminations": 2,
  "turnoverRate": "1.33%"
}
```

專案成本分析報表

```
GET /api/v1/reports/project/cost-analysis?projectId={id}
Response: {
  "project": "XX銀行專案",
  "budget": 10000000,
  "actualCost": 3500000,
  "utilizationRate": "35%",
  "profitMargin": "65%",
  "monthlyTrend": [
    {"month": "2025-11", "cost": 350000}
  ],
  "topCostMembers": [
    {"name": "張三", "hours": 120, "cost": 35000}
  ]
}
```

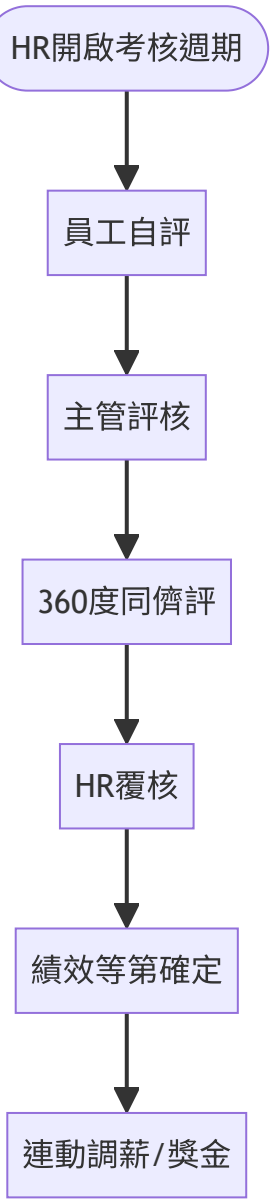
6.3 視覺化儀表板範例

高階主管儀表板配置:

```
{
  "dashboardName": "CEO Dashboard",
  "widgets": [
    {
      "type": "KPI_CARD",
      "title": "在職人數",
      "dataSource": "employee_count",
      "value": 150
    },
    {
      "type": "KPI_CARD",
      "title": "本月離職率",
      "dataSource": "turnover_rate",
      "value": "1.33%"
    },
    {
      "type": "LINE_CHART",
      "title": "月度人力成本趨勢",
      "dataSource": "monthly_labor_cost",
      "data": [...]
    },
    {
      "type": "PIE_CHART",
      "title": "部門人數分布",
      "dataSource": "headcount_by_department"
    }
  ]
}
```

7. 績效/招募/訓練服務 (08-10)

7.1 績效考核流程



7.2 招募漏斗分析

```
招募漏斗（某職缺）：
應徵者：100人
  ↓ 履歷篩選 (50%淘汰)
面試邀約：50人
  ↓ 面試 (60%淘汰)
Offer發放：20人
  ↓ 接受率 (70%)
到職：14人
  ↓ 試用期通過率 (90%)
留任：12人

招募效率指標：
- Time to Hire: 平均45天
- Cost per Hire: 平均50,000元
- Offer Acceptance Rate: 70%
```

7.3 訓練時數追蹤

```
法定訓練時數要求：
- 一般員工：12小時/年
- 管理職：20小時/年
- 新進員工：40小時（首年）

系統追蹤：
張三（一般員工）：
  已完成：8小時
  剩餘：4小時
  狀態：⚠️ 需注意（剩餘4個月）
```

---

8. 操作手冊補充

8.1 HR月度作業檢查清單

```
每月1日：
☐ 檢查上月考勤是否全部結算
☐ 確認所有請假/加班已審核

每月5日前：
☐ 執行月度薪資計算
☐ 檢查薪資計算異常
☐ 核准薪資
☐ 產生銀行薪轉檔
☐ 發送薪資單Email

每月10日：
☐ 檢查保險加退保申報
☐ 上傳勞保局e化系統

每月底：
☐ 檢查特休到期提醒
☐ 檢查合約到期提醒
☐ 檢查證照到期提醒
```

補充文件結束

涵蓋服務: 07工時管理、08-10績效/招募/訓練、11簽核流程、12通知、13文件、14報表分析
修訂日期: 2025-12-03
修訂人: SA